

Migration Shocks and Rent Growth in Supply-Constrained Housing Markets

Draft

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Abstract

The COVID-19 pandemic and the expansion of remote work changed the geography of household location choice in the United States. This paper studies whether pandemic-era population inflows were associated with faster local rent growth, and whether this relationship was stronger in housing markets with more restrictive supply conditions. I construct a county-year panel linking Zillow Observed Rent Index data, IRS county-to-county migration flows, American Community Survey controls, Census building permits, and land-use regulation measures. The preferred design classifies counties into high-, neutral-, and low-inflow groups based on pandemic-era net in-migration and estimates a binned difference-in-differences model with county and year fixed effects. I then test whether high-inflow counties experienced larger post-2020 rent increases in more zoning-restrictive or otherwise supply-constrained markets. The design shifts attention away from remote-work adoption itself, which is difficult to separate from broader pandemic shocks, and instead focuses on migration-driven housing demand as the proximate treatment. The empirical interpretation depends on parallel pre-trends and on defining migration exposure with a fixed, pre-specified pandemic window.

1 Introduction

The COVID-19 pandemic produced a large and uneven reshuffling of households across U.S. local housing markets. Remote work relaxed the link between workplace location and residence for many workers, allowing some households to leave expensive employment centers and move toward lower-cost, warmer, less dense, or higher-amenity areas. These moves were not evenly distributed across space. Some counties received sharp population inflows, while others experienced outflows or little change.

This paper asks whether these pandemic-era migration shocks contributed to local rent growth. The question is especially important in housing markets where supply cannot adjust quickly. If incoming households increase demand for rental housing, the rent response should be larger where vacancy rates are low, permitting is slow, housing-unit growth is limited, or zoning restrictions make new construction difficult. In contrast, more elastic markets should absorb inflows with smaller price effects.

The main research question is: did pandemic-era population inflows increase local rent growth, and were these effects stronger in supply-constrained housing markets?

This framing separates the project from a direct study of remote work adoption. Remote work is an important mechanism, but treating remote work itself as the treatment is empirically difficult because remote-work adoption coincided with many other shocks: pandemic health risk, fiscal transfers, interest-rate changes, labor-market restructuring, and local policy responses. I therefore treat migration inflows as the proximate local housing-demand shock and study how rents changed in places differentially exposed to that shock.

The contribution is threefold. First, the paper estimates the relationship between migration-driven demand and local rent growth using a county-year panel. Second, it tests whether rent responses were amplified by supply constraints, with zoning restrictiveness as the preferred institutional measure. Third, it clarifies the causal interpretation of pandemic migration studies by distinguishing remote work and amenities as background mechanisms from realized migration inflows as the main treatment exposure.

2 Background and Related Literature

This project relates to research on remote work, housing demand, and local housing supply. Mondragon and Wieland (2022) argue that the shift to remote work increased demand for housing, especially in areas attractive to remote workers. Howard (2023) studies the short- and long-run housing-market effects of remote work in the United States. Dingel and Neiman (2020) provide a widely used measure of work-from-home feasibility across occupations, which helps explain why some labor markets were more exposed to the remote-work transition.

The paper also connects to urban economics research on housing supply constraints. Standard models predict that a positive demand shock raises prices more in markets where housing supply is inelastic. Land-use regulation, geographic limits, low vacancy, and slow permitting can all reduce the ability of supply to respond. The Wharton Residential Land Use Regulatory Index provides one empirical measure of local regulatory restrictiveness and has been widely used to study housing supply constraints.

3 Data

3.1 Unit of Analysis

The unit of analysis is a county-year panel covering U.S. counties from the pre-pandemic period through the pandemic and early post-pandemic period. The baseline sample is intended to cover 2018 through 2023, subject to data availability.

3.2 Outcome: Rent Growth

The main outcome is annual local rent growth, measured as the change in log rental prices:

$$\Delta \log Rent_{ct} = \log Rent_{ct} - \log Rent_{c,t-1}.$$

The preferred rent measure is the Zillow Observed Rent Index (ZORI). ZORI is available monthly and is aggregated to the county-year level by averaging monthly index values within each county-year. A robustness check replaces ZORI with ACS median gross rent.

3.3 Treatment: Migration Shock

The main treatment exposure is pandemic-era population inflow. I construct net in-migration as a share of prior-year county population:

$$NetMigShare_{ct} = \frac{Inflow_{ct} - Outflow_{ct}}{Population_{c,t-1}}.$$

The baseline design classifies counties using cumulative pandemic-era net migration, preferably measured over 2020–2022:

$$PandemicShock_c = \sum_{t=2020}^{2022} NetMigShare_{ct}.$$

Counties in the top quartile of this distribution are classified as high-inflow counties. Counties in the bottom quartile are classified as low-inflow counties. Counties between these cutoffs form the neutral comparison group. Robustness checks use terciles, quintiles, and continuous migration measures.

3.4 Controls

Control variables include demographic, economic, and housing-market characteristics. These include population size, median household income, income growth, unemployment, renter share, vacancy rate, housing units, and permits per capita. The main sources are the American Community Survey and Census Building Permits Survey.

3.5 Supply Constraints

The preferred supply-constraint measure is zoning or land-use restrictiveness. The main specification uses a binary high-zoning indicator based on whether a county is above the median of a zoning restrictiveness index. Because zoning is slow-moving and may itself respond to long-run local economic conditions, the preferred measure should be pre-pandemic or time-invariant.

Additional supply-side measures include pre-pandemic rental vacancy rate, permits per capita, housing-unit growth, renter share, and alternative continuous zoning indices.

4 Empirical Strategy

4.1 Baseline Binned Difference-in-Differences

The baseline model compares rent growth before and after the pandemic shock across counties with different migration exposure:

$$\Delta \log Rent_{ct} = \beta_1 HighInflow_c \times Post_t + \beta_2 LowInflow_c \times Post_t + \alpha_c + \delta_t + X_{ct}\gamma + \epsilon_{ct}.$$

Here, α_c are county fixed effects, δ_t are year fixed effects, and X_{ct} is a vector of time-varying controls. The coefficient β_1 measures whether high-inflow counties experienced faster rent growth after 2020 relative to neutral counties, net of county and year fixed effects. The coefficient β_2 measures the analogous change for low-inflow counties.

The preferred interpretation is not that migration is randomly assigned. Instead, the model asks whether counties that received large pandemic-era inflows experienced differential rent growth after the shock, conditional on fixed county characteristics, common year shocks, and observed controls.

4.2 Supply-Constraint Heterogeneity

To test whether rent responses were larger in supply-constrained markets, I estimate:

$$\begin{aligned} \Delta \log Rent_{ct} = & \beta_1 HighInflow_c \times Post_t + \beta_2 HighInflow_c \times Post_t \times HighZoning_c \\ & + \beta_3 LowInflow_c \times Post_t + \alpha_c + \delta_t + X_{ct}\gamma + \epsilon_{ct}. \end{aligned}$$

The parameter of interest is β_2 . A positive β_2 would indicate that high-inflow counties experienced larger post-2020 rent growth when they were also zoning-restrictive. This is the main test of the supply-constraint mechanism.

4.3 Event-Study Specification

The key identifying assumption is that high-, neutral-, and low-inflow counties would have followed parallel rent-growth trends absent the pandemic migration shock. I assess this assumption with an event-study specification:

$$\Delta \log Rent_{ct} = \sum_{\tau \neq 2019} \theta_\tau HighInflow_c \times 1[t = \tau] + \alpha_c + \delta_t + X_{ct}\gamma + \epsilon_{ct}.$$

The omitted year is 2019. Coefficients for 2018 and other pre-period years should be close to zero if the comparison is credible. Large pre-trends would weaken the causal interpretation and suggest that high-inflow counties were already on different rent trajectories.

4.4 Instrumental Variables Extension

As an extension, I consider using pre-pandemic amenity and climate exposure interacted with the post-pandemic period as predictors of migration inflows:

$$ClimateAmenity_c \times Post_t \rightarrow MigrationShock_{ct} \rightarrow \Delta \log Rent_{c,t+1}.$$

This extension must be interpreted cautiously. Climate and amenities may affect rents directly, not only through migration, especially if preferences for space, weather, or local amenities changed during the pandemic. For that reason, I treat the IV design as supplementary rather than as the main identification strategy.

5 Identification Threats

The primary threat is endogenous migration. Counties with stronger labor markets, rising amenities, or better fiscal conditions may attract migrants and experience rent growth for reasons unrelated to a migration-demand shock. County fixed effects remove time-invariant differences, and year fixed effects remove common national shocks, but time-varying local shocks remain a concern.

A second threat is mechanical treatment definition. If high-inflow status is defined using the full post-period, including years in which rent growth is already responding, the treatment groups may reflect the same economic forces that drive the outcome. To reduce this problem, the treatment window should be fixed in advance, and robustness checks should show whether results change when the shock is defined using narrower windows.

A third threat is measurement error. IRS migration data are based on address changes observed in tax returns and do not capture all movers. In addition, the IRS changed its matching process beginning with the 2022–2023 migration files. The paper should therefore report whether results are robust to excluding 2023 or using only 2020–2022 migration exposure.

A fourth threat is rent-index coverage. ZORI is strongest in counties with sufficient rental listings and may underrepresent thin rural markets. The analysis should report sample coverage and test robustness using ACS median gross rent where possible.

6 Results

Table 2 reports the first-pass empirical results using the currently assembled ZORI and IRS migration panel. Column (1) estimates the binned difference-in-differences specification. The coefficient on $HighInflow_c \times Post_t$ is negative and statistically significant, while the coefficient on $LowInflow_c \times Post_t$ is also negative. This pattern suggests that the current high-, neutral-, and low-inflow binning should not yet be interpreted as a final causal estimate. In particular, the neutral comparison group may include counties with unusually strong post-pandemic rent growth, or the treatment window may be too contemporaneous with the outcome window.

Column (2) reports a continuous migration specification. In contrast to the binned model, the coefficient on net migration share is positive and statistically significant. This result is more consistent with the demand-shock hypothesis: counties with larger migration inflows experienced faster rent growth. However, because realized migration remains endogenous, this estimate should be interpreted as conditional evidence rather than a fully causal effect.

Table 1: Summary Statistics by Migration-Shock Group

Group	Counties	Mean rent growth	Mean net migration share	Mean rent
High inflow	189	0.067	0.023	1412
Neutral	129	0.061	0.000	1244
Low inflow	189	0.053	-0.011	1520

Table 2: Migration Shocks and County Rent Growth

	(1) Binned DiD	(2) Continuous migration
High inflow $\times$ Post	-0.0046 (0.0029)	
Low inflow $\times$ Post	-0.0064*** (0.0022)	
Net migration share		1.7704*** (0.3397)
County fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	4,668	1,740
R^2	0.605	0.709

Notes: The outcome is annual change in log ZORI rent. The binned model classifies high- and low-inflow counties using the top and bottom quartiles of cumulative 2020–2022 net migration share; neutral counties are those with cumulative net migration share within 0.5 percentage points of zero. Standard errors are clustered by county. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The divergence between the binned and continuous specifications is useful diagnostically. The five-bin model in Table 3 also does not produce a monotonic positive DiD pattern across migration-shock groups. This indicates that the next stage of the analysis should focus on pre-trend event studies, alternative bin definitions, and lagged timing in which migration shocks are measured before subsequent rent growth. The zoning heterogeneity test is not yet reported because the county-level zoning restrictiveness file has not been merged into the analysis panel.

7 Robustness Checks

The paper should include the following robustness exercises: vary migration bins using terciles, quartiles, and quintiles; estimate continuous specifications using net in-migration share; replace ZORI rent growth with ACS median gross rent growth; exclude superstar or large coastal counties;

Table 3: Five-Bin Migration Shock Difference-in-Differences

Outcome: annual log rent growth	
Q1 lowest migration $\times$ Post	-0.0086*** (0.0029)
Q2 $\times$ Post	-0.0025 (0.0031)
Q4 $\times$ Post	-0.0045 (0.0029)
Q5 highest migration $\times$ Post	-0.0085*** (0.0026)
Reference group	Q3 middle migration quintile
County fixed effects	Yes
Year fixed effects	Yes
Observations	6,708
R^2	0.623

Notes: Counties are divided into five bins using cumulative 2020–2022 net migration share. Q3 is omitted as the reference group. Standard errors are clustered by county. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

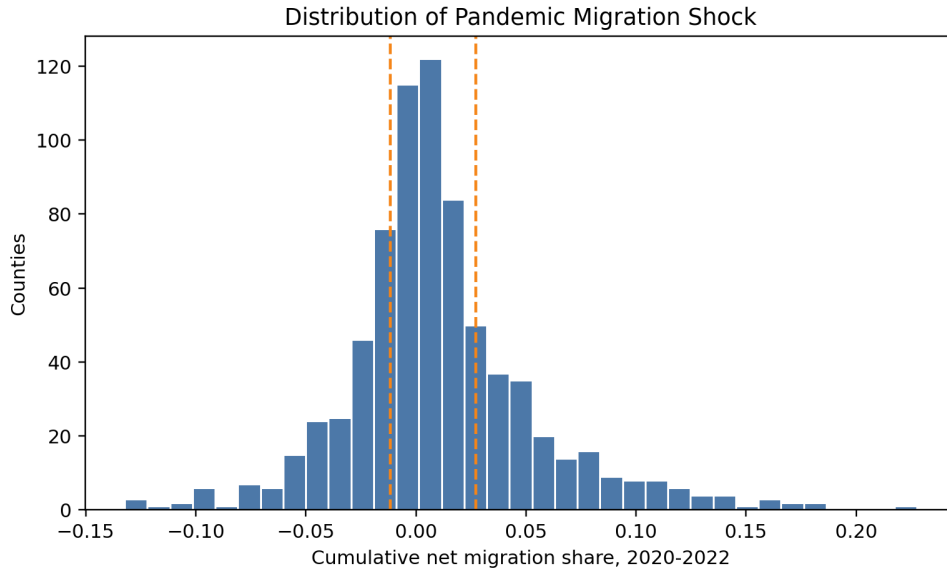


Figure 1: Distribution of Pandemic Migration Shock

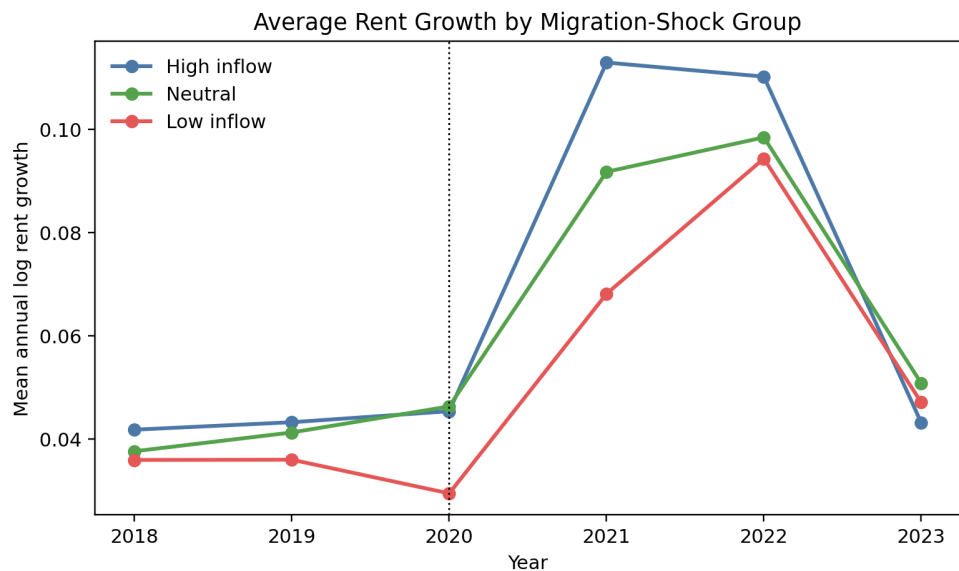


Figure 2: Average Rent Growth by Migration-Shock Group

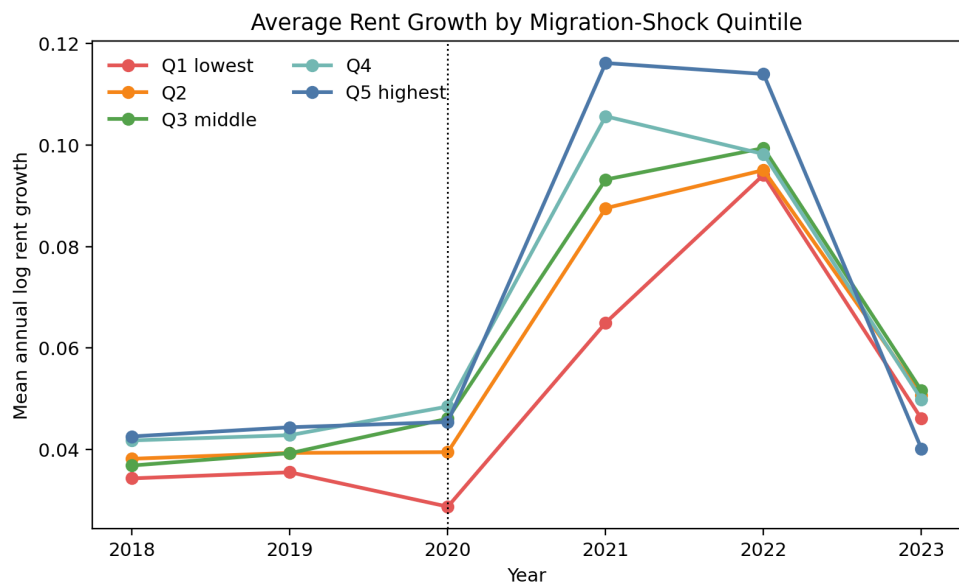


Figure 3: Average Rent Growth by Migration-Shock Quintile

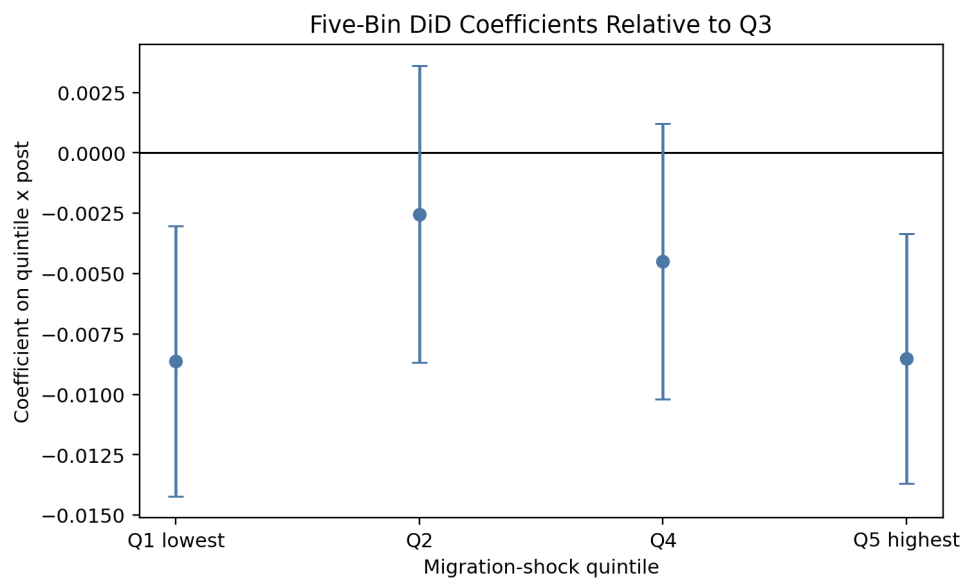


Figure 4: Five-Bin DiD Coefficients Relative to the Middle Quintile

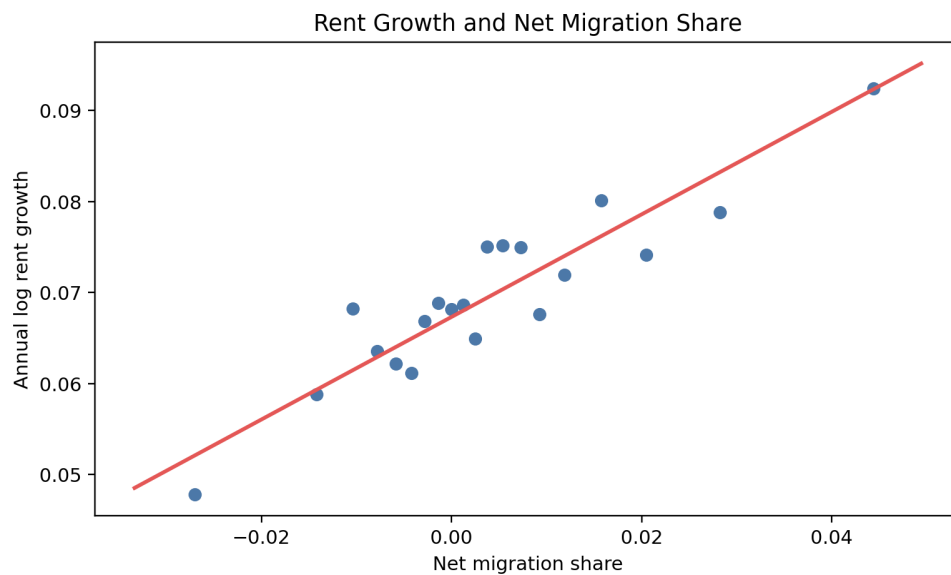


Figure 5: Rent Growth and Net Migration Share

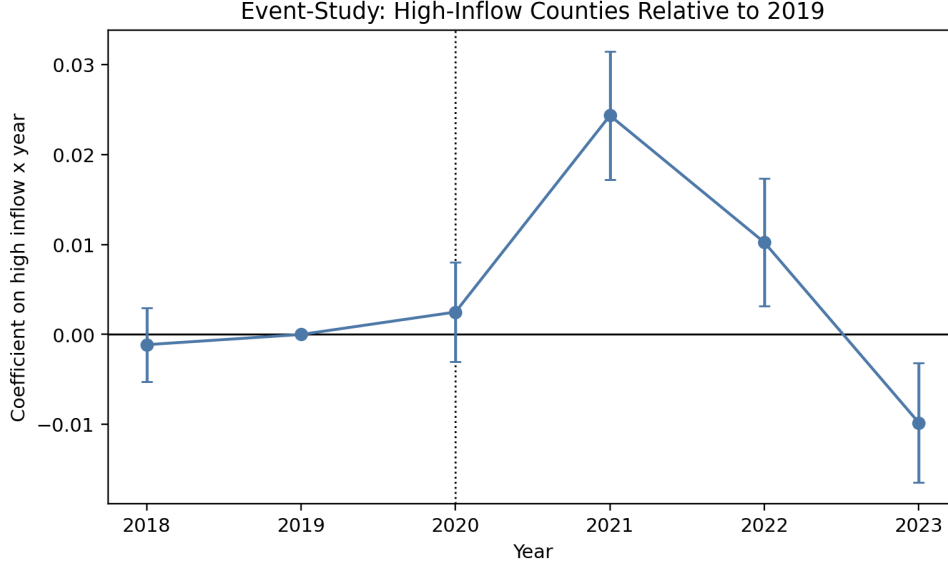


Figure 6: Event-Study Coefficients for High-Inflow Counties

aggregate counties to metropolitan areas where appropriate; exclude 2023 migration data because of the IRS matching-process change; estimate placebo DiD models using only pre-pandemic years; replace zoning restrictiveness with vacancy rate, permits per capita, and housing-unit growth; use continuous zoning indices rather than a binary high-zoning indicator; and weight regressions by county population and compare with unweighted estimates.

8 Expected Contribution

This paper contributes to the literature by reframing pandemic remote-work housing effects as a migration-driven local demand shock. This distinction matters for causal inference. Remote work was part of a broader macroeconomic and social shock, while county-level migration inflows provide a more direct measure of local housing-demand pressure. By interacting migration shocks with supply constraints, the paper identifies which types of markets were most vulnerable to rent pressure during the pandemic reshuffling.

The policy relevance is direct. If migration shocks have larger rent effects in zoning-restrictive markets, then affordability pressure after demand shocks is not only a function of how many people move, but also of whether local institutions allow housing supply to respond.

9 Conclusion

Pandemic-era migration reshaped local housing demand across U.S. counties. This paper proposes a county-year empirical design to estimate whether high-inflow counties experienced faster rent growth after 2020 and whether this effect was amplified in supply-constrained markets. The

strongest version of the design uses fixed migration-shock bins, county and year fixed effects, pre-trend tests, and pre-pandemic measures of zoning or supply constraints.

The central empirical prediction is that migration inflows should raise rents more in counties where housing supply is less elastic. If supported by the data, the findings would suggest that pandemic migration contributed to local rental pressure and that supply restrictions played an important role in translating demand shocks into affordability shocks.

References

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